

Problem 11

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Problem 1 The total number of people, N who have contracted a common cold by a time t days after its outbreak on an island is given by $N = N(t) = \frac{200000}{1 + 100e^{-0.1t}}, t \geq 0$.

- (a) Evaluate and interpret the limit $\lim_{t \rightarrow \infty} N(t)$

Explanation. $\lim_{t \rightarrow \infty} N(t) = \lim_{t \rightarrow \infty} \frac{200000}{1 + 100e^{-0.1t}} = 200000$. In the long run, 200000 people will get the cold.

- (b) How long will it take for the number of people who have contracted the cold to reach 40,000?

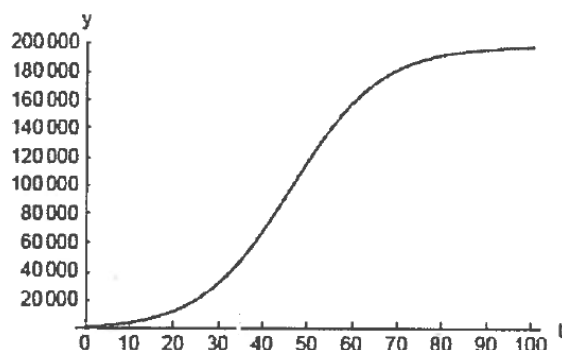
Explanation.

$$\begin{aligned}
 N(t) &= 40000 \\
 \frac{200000}{1 + 100e^{-0.1t}} &= 40000 \\
 200000 &= 40000(1 + 100e^{-0.1t}) \\
 1 + 100e^{-0.1t} &= \frac{200000}{40000} \\
 100e^{-0.1t} &= 5 - 1 \\
 e^{-0.1t} &= \frac{4}{100} \\
 -0.1t &= \ln(.04) \\
 t &= \frac{\ln(0.04)}{-0.1} = 32.1888
 \end{aligned}$$

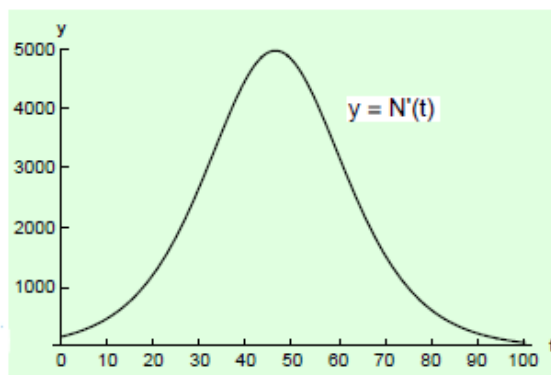
It will take 33 days for the number of people who have contracted the cold to reach 40,000.

- (c) The graph of the function N on the interval $[0, 100]$ is given below. Sketch (as best you can) the graph of its derivative, $N'(t)$.

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Explanation.



- (d) Calculate $N'(t)$. What does $N'(t)$ represent?

Explanation.
$$N'(t) = \frac{-200000(10)e^{-0.1t}}{(1 + 100e^{-0.1t})^2} = \frac{2000000e^{-0.1t}}{(1 + 100e^{-0.1t})^2}.$$

$N'(t)$ represents the instantaneous growth rate of the number of people who have contracted the cold at the time t .

- (e) Evaluate and interpret the limit $\lim_{t \rightarrow \infty} N'(t)$

Explanation.
$$\lim_{t \rightarrow \infty} N'(t) = \lim_{t \rightarrow \infty} \frac{2000000e^{-0.1t}}{(1 + 100e^{-0.1t})^2} = \frac{2000000(0)}{(1 + 100(0))^2} = \frac{0}{1} = 0.$$

In the long run, the number of people who have contracted the cold will stabilize. There will be no growth in the long run.

- (f) Find the average growth rate of the number of people who have contracted the disease during the time interval $[5, 6]$ (or during the sixth day after the outbreak).

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Explanation. $AVRG = \frac{N(6) - N(5)}{6 - 5} = N(6) - N(5) = \frac{200000}{1 + 100e^{-0.1(6)}} - \frac{200000}{1 + 100e^{-0.1(5)}} \approx 335.1$

- (g) Find the instantaneous growth rate of the number of people who have contracted the disease for $t = 5$ (round to a whole number).

Explanation. $N'(5) = \frac{2000000e^{-0.1(5)}}{(1 + 100e^{-0.1(5)})^2} = 319$ people per day